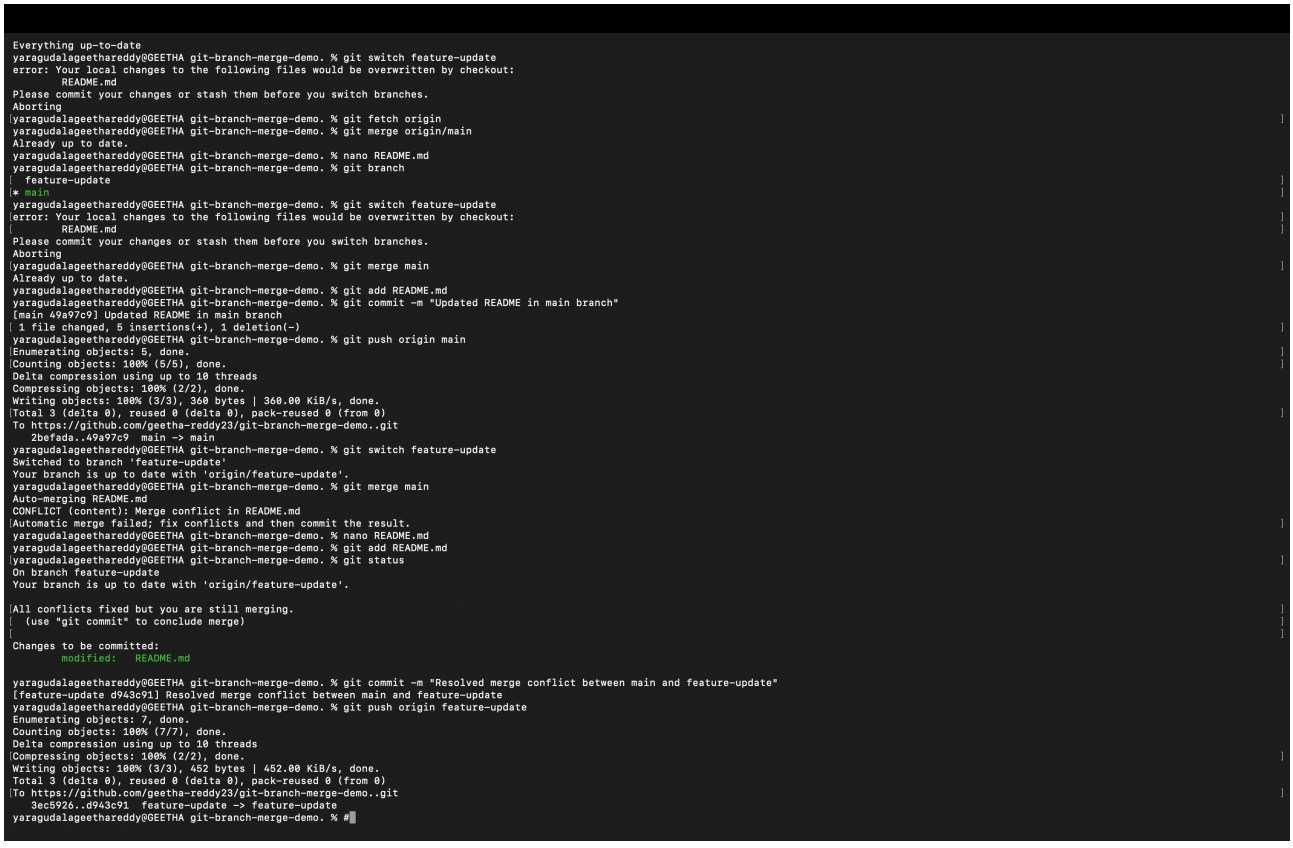
Experiment 10: Git Branching and Merge Conflict Resolution

Aim:

Demonstrate how to work with Git branches and resolve merge conflicts when collaborating with others.

Tasks:

1. Clone a shared repository to your local machine.
2. Create a new branch and switch to it.
3. Make changes to a file (e.g., update a README or modify code in a specific function).
4. Commit your changes.
5. Push the changes to the remote repository.
6. Before merging, pull the latest changes from the main branch.
7. Switch back to your feature branch.
8. Merge the main branch into your feature branch.
9. If there are conflicts, resolve them manually by editing the conflicting files. After resolving conflicts, mark the conflicts as resolved.
10. Commit the resolved merge.
11. Push your feature branch with the merged changes to GitHub and create a pull request.
12. Submit a summary of the steps you performed, the conflicts you encountered, and how you resolved them.

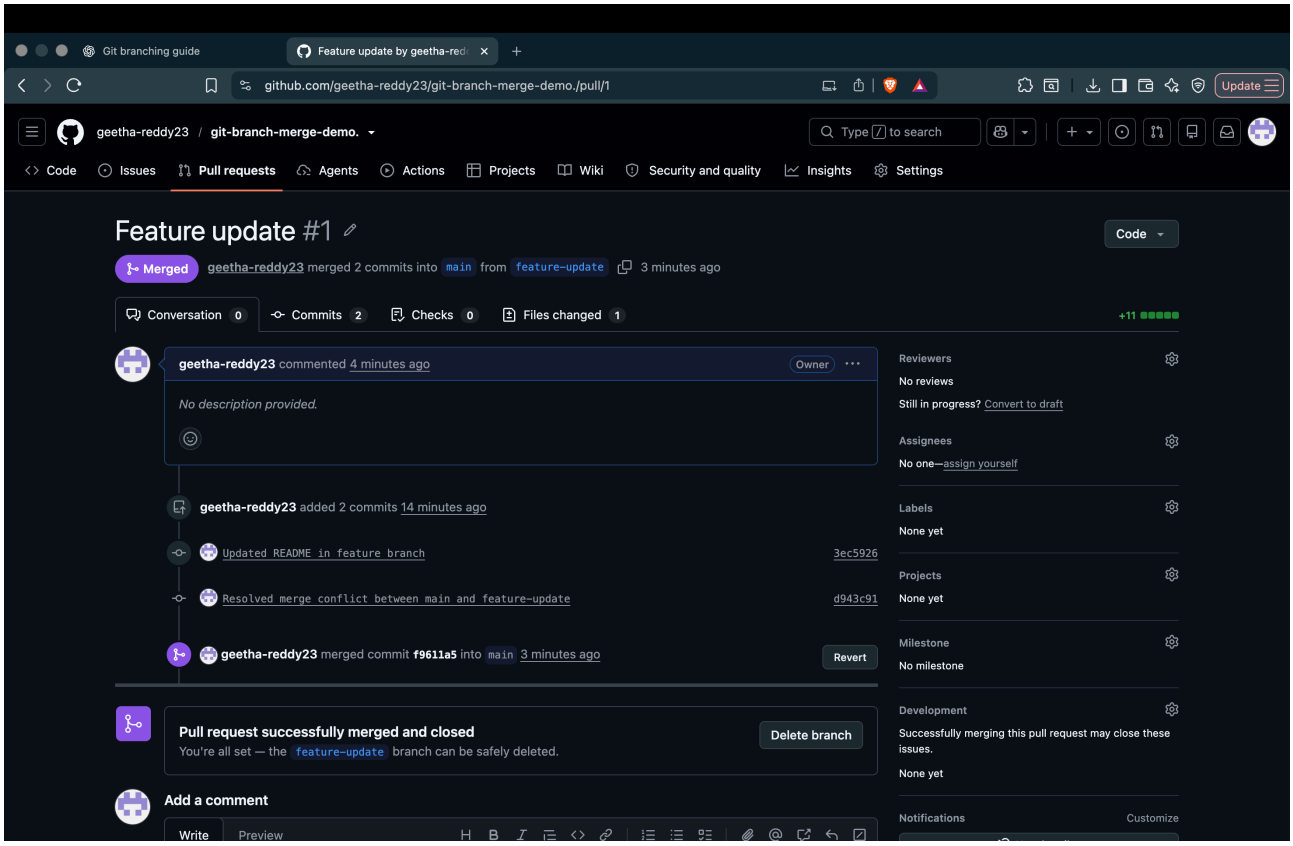


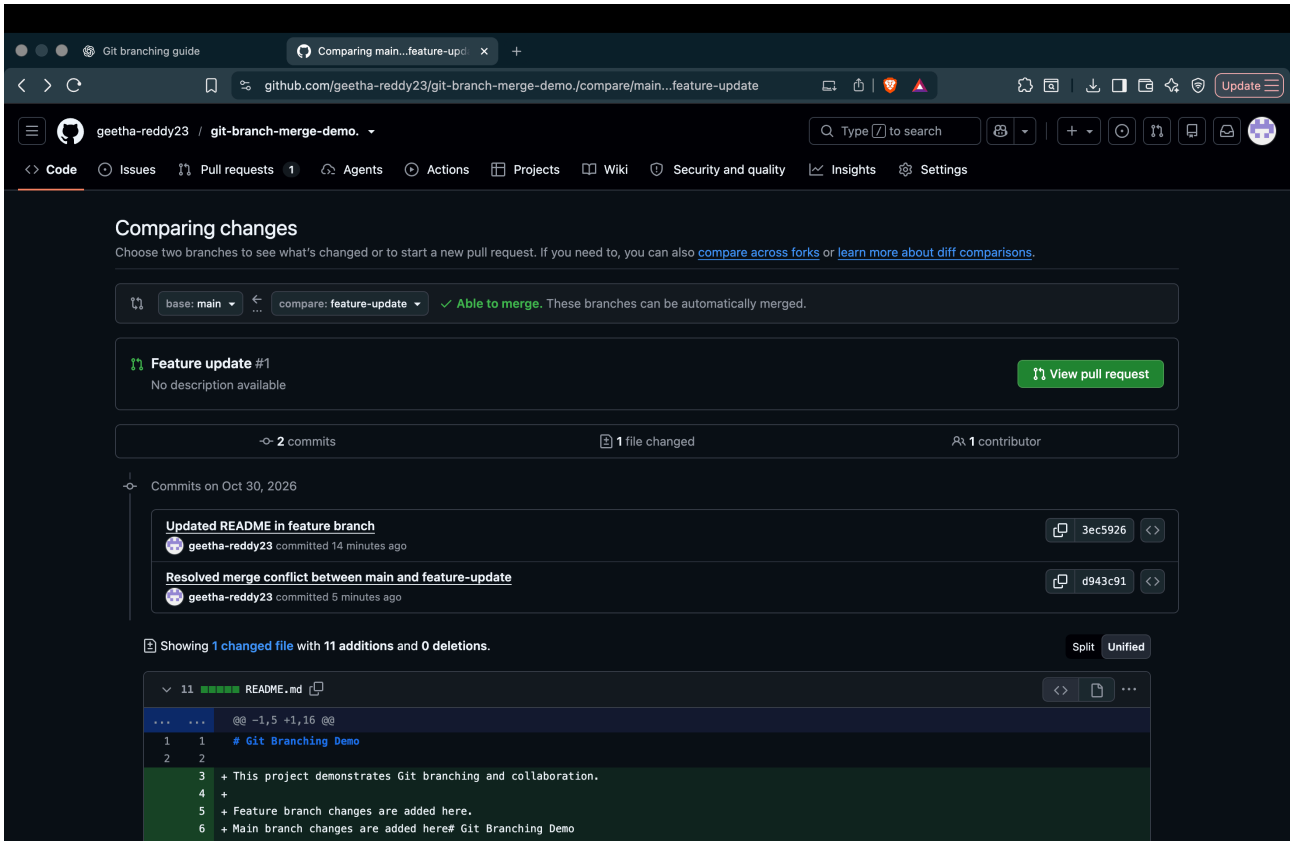
```
Last login: Mon Oct 26 14:39:10 on ttys018
yaragudalageethareddy@GEETHA ~ % git clone https://github.com/geetha-reddy23/git-branch-merge-demo..git
Cloning into 'git-branch-merge-demo.'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
yaragudalageethareddy@GEETHA ~ % https://github.com/geetha-reddy23/git-branch-merge-demo..git
zsh: no such file or directory: https://github.com/geetha-reddy23/git-branch-merge-demo..git
yaragudalageethareddy@GEETHA ~ % cd git-branch-merge-demo.
yaragudalageethareddy@GEETHA git-branch-merge-demo. % ls
README.md
yaragudalageethareddy@GEETHA git-branch-merge-demo. % ls
README.md
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git branch
* main
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git branch feature-update
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git switch feature-update
Switched to branch 'feature-update'
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git branch
* feature-update
  main
yaragudalageethareddy@GEETHA git-branch-merge-demo. % nano README.md
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git status
On branch feature-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git add README.md
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git commit -m "Updated README in feature branch"
[feature-update 3ec5926] Updated README in feature branch
 1 file changed, 5 insertions(+), 1 deletion(-)
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git push -u origin feature-update
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 357 bytes | 357.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-update' on GitHub by visiting:
remote:      https://github.com/geetha-reddy23/git-branch-merge-demo./pull/new/feature-update
remote:
To https://github.com/geetha-reddy23/git-branch-merge-demo..git
 * [new branch]      feature-update -> feature-update
branch 'feature-update' set up to track 'origin/feature-update'.
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
yaragudalageethareddy@GEETHA git-branch-merge-demo. % nano README.md
yaragudalageethareddy@GEETHA git-branch-merge-demo. % # Git Branching Demo

This project demonstrates Git collaboration.

Main branch changes are added here.
zsh: command not found: #
zsh: command not found: This
zsh: command not found: Main
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git push origin main
Everything up-to-date
yaragudalageethareddy@GEETHA git-branch-merge-demo. % git switch feature-update
```




Git branching guide
Comparing main...feature-upd
github.com/geetha-reddy23/git-branch-merge-demo./compare/main...feature-update
Update
geetha-reddy23 / git-branch-merge-demo.
Type / to search
Code
Issues
Pull requests 1
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about diff comparisons.
base: main
compare: feature-update
Able to merge. These branches can be automatically merged.
Feature update #1
No description available
View pull request
2 commits
1 file changed
1 contributor
Commits on Oct 30, 2026
Updated README in feature branch
geetha-reddy23 committed 14 minutes ago
3ec5926
Resolved merge conflict between main and feature-update
geetha-reddy23 committed 5 minutes ago
d943c91
Showing 1 changed file with 11 additions and 0 deletions.
Split
Unified
11 README.md
@@ -1,5 +1,16 @@
Git Branching Demo
+ This project demonstrates Git branching and collaboration.
+
+ Feature branch changes are added here.
+ Main branch changes are added here# Git Branching Demo